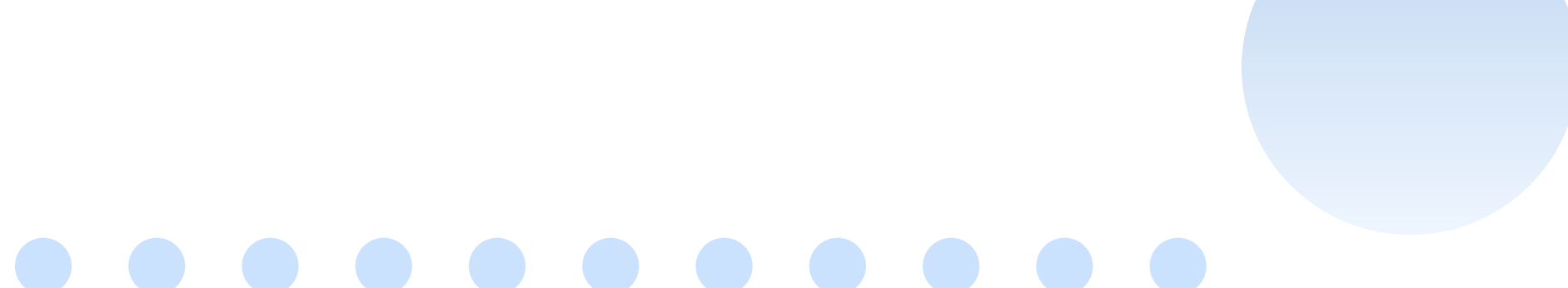


Extended Reality Products + in Surgical Care

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Introduction +

Problem

How can extended reality products transform surgical care?

Surgical Market Size

2024 – **\$742.93 Billion**

Key Factors

- Tech advancements are driving market increase
- **Key Players:**
 - **Hospitals:** surgeons, patients, government, insurance
 - **Medical Device Companies:** Medtronic, Stryker, Abbott, Augmedics

Extended Reality (XR)

- Augmented/Virtual Reality
- **Applications**
 - Improve training
 - Facilitate preparation and the operation
 - Make operations personalized and less invasive
 - Support patient rehabilitation

Impact of XR Devices

Surgical Procedure Costs

	Cost per Case (\$)
Minor procedure item and task break down	
Minor procedure supplies	25.71
Variables	0.66
Staff	21.63
Room	5.83
Instrument depreciation	0.53
Blogging/Web site	9.50
Total	63.86
Major procedure item and task break down	
Preoperative time	90.60
Perioperative time	265.80
Postoperative recovery supplies	2.88
Sterilization supplies	3.68
Anesthesia administration	312.22
Office administration	156.94
Laundry	11.35
Room cost	98.00
Blogging/Web site	9.50
Total	950.97

This table summarizes total office costs for minor and major procedures, taking into account factors like administration time, Web site development, and rent per square foot of the rooms used.

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XR Products



100 + AR/VR FDA-listed devices:

Medtronic: StealthStation S8 platform for 3D imaging and guiding approach

Stryker: Target Guided Surgery System for enhanced endoscopy usage

Apple Vision Pro: Improved visualization and ergonomics during the operation

Increase in Technological Costs
(\$200-\$200,000)

Decrease Operative Time by
10-15%

Reduce Training Costs by
30-40%

Opportunities

Efficiency-Driven

Focus on utilizing XR to optimize current surgical workflows.

Revenue Impact

- Reduce preoperative time and operating room time/turnover without increasing staff
- Overall savings defends investment

Strategy/Implementation

- Hospitals continue to integrate full XR suites
- Policymakers protect XR data and make devices more accessible

Patient-Driven

Focus on utilizing XR to improve patient outcomes.

Revenue Impact

- Decrease negative efficiency adjustments for physician fee schedule
- Reduce patient complications and readmissions

Strategy/Implementation

- New CMS ACCESS Reimbursement codes for Outcome-Aligned Payments

Training-Focused

Focus on utilizing XR to fully replace cadaver training labs and improve training efficiency.

Revenue Impact

- Reduce supply and biohazard management costs for training
- Reduce training time and costs

Strategy/Implementation

- Changing Graduate Medical Education and medical board approved residency programs
- Develop specialized training hubs